Exponential Applications 2

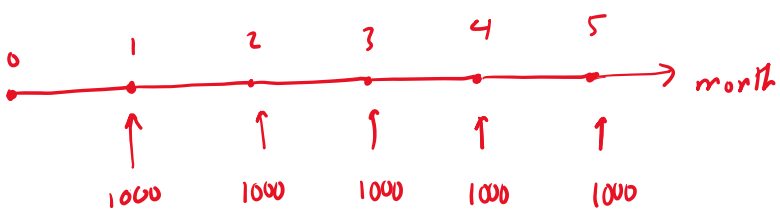
1. Annuity

An **annuity** is a financial contract that provides a series of regular, fixed payments over a specific period.

Depending on your goal, an annuity functions in one of two ways:

- **Accumulation Phase (Building):** You make regular contributions to an account that grows with interest over time to reach a future goal.
- **Distribution Phase (Receiving):** You start with a lump sum that is paid out to you in periodic installments until the balance reaches zero.

1.1. Building An Annuity - Future Value



(A) Invest / saving $1000 in the end of every month for 5 months.

(A) How much the total investment is worth after 5 months if the interest rate is 10%

The Formula

$$FV = P \times \left[\frac{(1+r)^n - 1}{r} \right]$$

Where:

- FV: The Future Value (the total amount you want to have).
- P: The Periodic payment (the amount you deposit each period).
- r: The interest rate per period (expressed as a decimal).
- n: The total number of periods (e.g., total years or months).

In the example: $P = 1000$

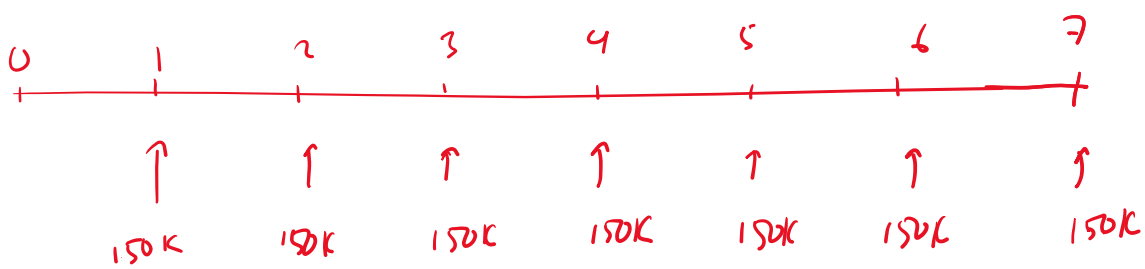
$n = 5$

$r = 10\%$

$$FV = 1000 \cdot \left[\frac{(1+.1)^5 - 1}{.1} \right] = 6105.1$$

Example 1: A rookie player in the NFL just signed his first 7-year contract. To prepare for his future, he deposits $150,000 at the end of each year for 7 years in an account with 3.1% compounded annually.

a. How much will he have in his account after 7 years?



$$\Rightarrow P = F \quad 150{,}000 \cdot \left[\frac{(1+.031)^7 - 1}{.031} \right] = 1{,}152{,}854.59$$

b. How much interest was earned over the 7 years?

without the investment: total amount: $150K \times 7$

$= 1{,}050{,}000$

102,854

Interest earned: 1,152,854 − 1,050,000 =

Example 2: When a worker first starts out, they do not have a lot of extra money to put into savings. They are able to invest $400 per month at 2.3% compounded monthly for the first 10 years. After those 10 years, they are making more money and can afford to put more money into an account each month. The worker transfers all of the money they have from their first account into a new account with 2.89% interest. They put $700 per month into this account for 20 years. How much money will be in the account at the end of the 30 years?

1.2. Receiving An Annuity- Present Value

The Formula

$$PV = P \times \left[\frac{1-(1+r)^{-n}}{r}\right]$$

Where:

- PV: The Present Value (the lump sum you start with today).
- P: The Periodic payment (the amount you receive each period).
- r: The interest rate per period (as a decimal).
- n: The total number of periods.

Example 3: A wealthy family member has funded an annuity that will pay you $1500 at the end of each year for six years. If the interest rate is 8%, compounded annually, find the present value of the annuity.

Example 4: To supplement his pension in the early years of his retirement, Ryan plans to use $124,500 of his savings as an ordinary annuity that will make monthly payments to him for 20 years. If the interest rate is 2.8%, how much will each payment be?

2. Loans and Amortization

Amortization is the process of paying off a debt over a set period through a series of regular, fixed payments.

$$M = L \times \left[\frac{r(1+r)^n}{(1+r)^n - 1} \right]$$

Variable Guide

- M: Monthly Payment (The amount you pay each period).
- L: Loan Amount (The total principal you are borrowing today).
- r: Monthly Interest Rate (Annual Rate / 12).
- n: Total Number of Payments (Loan Term in years $\times$ 12).

Example 5: The average used car price is $32,000. The average interest rate is currently 6.39% for a 60-month loan (5 years). If you have $5,000 for a down payment, what can you expect the monthly payment to be? How much will you pay in total for the car?

Example 6: The average rate for a 30-year fixed mortgage is 7.5%. Assume a down payment of 20% on a home purchase of $315,000.

a. Find the monthly payments needed.

b. How much will be paid in total over the life of the loan?

Example 7: In 2023, the average amount of student loan debt for a bachelor's degree recipient at a private university was $54,921. Assume an interest rate of 6.87%

a. Find the monthly payment and calculate the total interest paid over the course of the loan for a 10-year payment plan.

b. Find the monthly payment and calculate the total interest paid over the course of the loan for a 20-year payment plan.